Experiment No : 5

Title: Implement Stack and Queue using arrays in C.

Name of Student: Shamiksha Chaudhari
Class: SY **Div:** SY1 **PRN No:** B25ET1005 **Batch:** A1

Program:

Part 1
Stack using array

```
#include <stdio.h>
#define MAX 5
int stack[MAX];
int top = -1;
void push(int value) {
if (top == MAX - 1) {
printf("Stack Overflow! Cannot push %d.\n", value);
} else {
stack[++top] = value;
printf("Pushed %d to stack.\n", value);
}
}
void pop() {
if (top == -1) {
printf("Stack Underflow! Cannot pop.\n");
} else {
printf("Popped %d from stack.\n", stack[top--]);
}
}
void peek() {
if (top == -1) {
printf("Stack is empty.\n");
```

```
} else {
printf("Top element is %d.\n", stack[top]);
}
}
void display() {
if (top == -1) {
printf("Stack is empty.\n");
} else {
printf("Stack elements: ");
for (int i = 0; i <= top; i++) {
printf("%d ", stack[i]);
}
printf("\n");
}
}
int main() {
push(1);
push(2);
push(3);
push(4);
push(5);
push(6); // Stack Overflow
display();
peek();
pop();
display();
return 0;
```

Output

```
Pushed 1 to stack.
Pushed 2 to stack.
Pushed 3 to stack.
Pushed 4 to stack.
Pushed 5 to stack.
Stack Overflow! Cannot push 6.
Stack elements: 1 2 3 4 5
top element is 5 from stack.
Popped 5 from stack.
Stack elements: 1 2 3 4
```

PART2

Queue using array

```
#include <stdio.h>
#define SIZE 5 // maximum size of queue
int queue[SIZE];
int front = -1, rear = -1;
// Enqueue (insert element)
void enqueue(int value) {
if (rear == SIZE - 1) {
printf("Queue is Full!\n");
} else {
if (front == -1) front = 0;
queue[++rear] = value;
printf("%d inserted\n", value);
}
}
// Dequeue (delete element)
```

```
void dequeue() {
if (front == -1 || front > rear) {
printf("Queue is Empty!\n");
} else {
printf("%d deleted\n", queue[front++]);
}
}
// Display Queue
void display() {
if (front == -1 || front > rear) {
printf("Queue is Empty!\n");
} else {
printf("Queue: ");
for (int i = front; i <= rear; i++) {
printf("%d ", queue[i]);
}
printf("\n");
}
}
int main() {
int choice, value;
while (1) {
printf("\n1. Enqueue\n2. Dequeue\n3. Display\n4. Exit\n");
printf("Enter choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
printf("Enter value: ");
scanf("%d", &value);
enqueue(value);
break;
case 2:
```

```
dequeue();
break;
case 3:
display();
break;
case 4:
return 0;
default:
printf("Invalid choice!\n");
}
}
}
```